

Joseph C. Cuthbert

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EDUCATION

Georgia Institute of Technology

M.S. Computational Science and Engineering

Berry College

B.S. Computer Science and Mathematics

Cumulative GPA: 3.76/4.00

NCAA Men's Lacrosse Team, National College Athlete Honor Society

Atlanta, Georgia

August 2025 — December 2026 (Expected)

Mount Berry, Georgia

August 2021 — May 2025

EXPERIENCE

Sensors Directorate Intern

University of Dayton, NASIC, AFRL

May 2026 — August 2026

Dayton, Ohio

- Studied the robustness of CNN classification models by exploring the effect of noise attacks on SAR images
- Implemented a ResNet-18 model for transfer learning on SAR data and utilized high performance computing tools and PyTorch Lightning to design efficient and reproducible experiments
- Improved CNN robustness to noisy data by adding isolated targets into the training set, increasing classification accuracy by up to 66.7% under heavy noise

Machine Learning Research Assistant

Berry College Department of Computer Science

June 2023 — January 2025

Mount Berry, Georgia

- Co-authored a peer-reviewed paper published at FLAIRS-39 (2026) investigating whether team synergy can be extracted from team composition alone for sport outcome prediction
- Designed a Siamese Transformer model treating players as tokens, capturing intra-team interactions via shared self-attention, and predicting win probabilities from coordinate-wise team representation differences

Data and AI Intern

Premise Health

May 2024 — August 2024

Nashville, Tennessee

- Developed a predictive model for the number of visits across hundreds of the company's health centers and used Databricks NLP capabilities to allow no-code users to access our forecasts using generated SQL queries
- Assessed the company's new AI infrastructure to help determine where ML could provide the most value and what tools were essential to large-scale ML projects

PROJECTS

Local Search for Minimum Vertex Cover

Graduate Algorithms Project

January 2026 — April 2026

- Designed and implemented a hybrid NuMVC / Simulated-Annealing local search algorithm for the MVC problem in C
- Benchmarked on DIMACS graphs, reaching the optimal cover size on 10/10 seeded runs of a 500-vertex instance in an average of 0.34 seconds, and on a harder 1,000-vertex instance raised success from 28/60 to 59/60 runs via a perturbation mechanism, cutting mean time-to-target from 21.56s to 7.64s

Mechanistic Interpretability Research Project

Senior Capstone Project

January 2025 — May 2025

- Used TransformerLens to implement direct logit attribution and two corruption activation patching experiments to identify attention heads in GPT-2-small responsible for acronym formation
- Built an automated circuit-discovery (Edge Attribution Patching) pipeline that independently converged on the same four heads, and confirmed the finding by replicating an independently published paper's results

LEADERSHIP

Assistant Lacrosse Coach

North Atlanta High School

January 2026 — May 2026

Atlanta, Georgia

- Designed drills and offensive schemes for 30 athletes; team qualified for GHSA state tournament

Computer Science Teaching Assistant

Berry College Department of Computer Science

August 2023 — December 2023

Mount Berry, Georgia

- Led weekly labs and office hours for 40 students in functional programming; delivered a lecture on graph algorithms